

Chapter 8

Spherical symmetry and coordinates

8.1 Grounded conducting sphere

Consider a grounded conducting sphere of radius R , with a point charge q on the z -axis at distance z from the center, as depicted in the Figure. We want the potential ϕ outside the sphere, solving $\nabla^2\phi = -4\pi\rho$ with $\phi = 0$ on the surface. We use the image method: place an image charge q' inside the

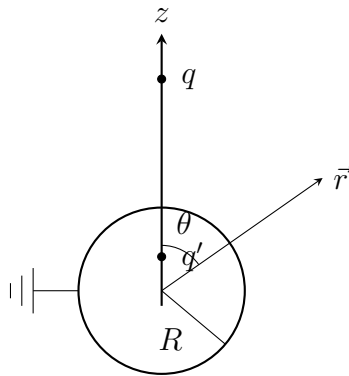


Figure 8.1: A grounded conducting sphere of radius R with a point charge q on the z -axis at distance z from the center; the image charge q' sits inside the sphere at distance z' along the same axis.

sphere at distance z' along the axis, and write

$$\phi = \frac{q}{|\vec{r} - z \hat{z}|} + \frac{q'}{|\vec{r} - z' \hat{z}|} \quad (8.1)$$

The boundary condition is $\phi = 0$ at $r = R$

$$\phi = \frac{q}{R \left(1 + \frac{z^2}{R^2} - \frac{2z}{R} \cos \theta\right)^{1/2}} + \frac{q'}{z' \left(1 + \frac{R^2}{z'^2} - \frac{2R}{z'} \cos \theta\right)^{1/2}} = 0 \quad (8.2)$$

which requires

$$\frac{q}{R} = -\frac{q'}{z'} \quad , \quad \frac{z}{R} = \frac{R}{z'} \quad (8.3)$$

that is

$$z' = \frac{R^2}{z} \quad , \quad q' = -\frac{R}{z} q \quad (8.4)$$

The surface charge density σ follows from the radial electric field at the surface

$$E_r = 4\pi\sigma = -\left.\frac{\partial\phi}{\partial r}\right|_{r=R} \quad , \quad \int \sigma da = q' \quad (8.5)$$

i.e. the total induced charge on the sphere equals the image charge. The force on q is

$$\begin{aligned} F_z &= \frac{q q'}{(z - z')^2} = -\frac{q^2 R/z}{z^2 \left(1 - \frac{R^2}{z^2}\right)^2} \\ &\sim -\frac{q^2}{4(z - R)^2} < 0 \quad \text{as } z \rightarrow R \end{aligned} \quad (8.6)$$

an attraction near the surface – a work function that sticks charges to the surface despite repulsion.

8.2 Ungrounded conducting sphere with charge Q

Next, let the sphere be isolated, carrying total charge Q , with the point charge q outside as before; we again want ϕ . The extra charge $Q - q'$ must be uniformly distributed, since there is no force from the pair q and q' at $r = R$. Hence

$$\phi \rightarrow \phi + \frac{Q - q'}{r} \quad (8.7)$$

There is still an attractive force near $z \sim R$.

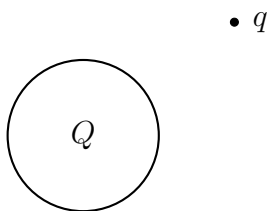


Figure 8.2: An isolated conducting sphere carrying total charge Q , with a point charge q outside.

8.3 Conducting sphere at potential ϕ_0

If instead the sphere is held at fixed potential ϕ_0 , as in the Figure, the

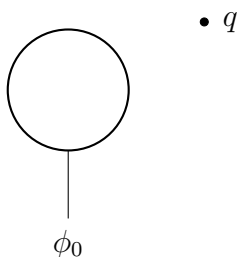


Figure 8.3: A conducting sphere held at fixed potential ϕ_0 , with a point charge q outside.

potential becomes

$$\phi \rightarrow \phi + \frac{\phi_0 R}{r} \quad (8.8)$$

8.4 Conducting sphere in a uniform electric field

Surround the sphere with a charge $-q$ at distance b above the center and a charge $+q$ at distance b below, as in the Figure, and take

$$E_0 = \frac{2q}{b^2} = \text{fixed} \quad \text{while} \quad q, b \rightarrow \infty \quad (8.9)$$

The potential ϕ is then given by the four charges $-q$, $+q$, q' , $-q'$.

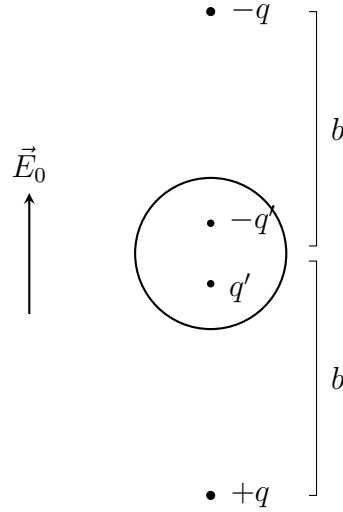


Figure 8.4: A conducting sphere in an asymptotically uniform electric field $\vec{E}_0$, built from charges $\mp q$ at distances b above and below the center, together with their images $\mp q'$ inside the sphere.

8.5 General Green's function with Dirichlet conditions

For the spherical geometry, the Dirichlet Green's function G_D follows from the previous example (8.4) by the replacements

$$z \rightarrow r' \quad , \quad z' \rightarrow \frac{R^2}{r'} \quad , \quad q \rightarrow 1 \quad , \quad q' \rightarrow -\frac{R}{r'} \quad (8.10)$$

giving

$$G_D(r, r') = \frac{1}{|\vec{r} - \vec{r}'|} - \frac{R}{r' \left| \frac{R^2}{r'^2} \vec{r}' - \vec{r} \right|} \quad (8.11)$$

that is

$$G_D(r, r') = \frac{1}{\sqrt{r^2 + r'^2 - 2 r r' \cos \alpha}} - \frac{1}{\sqrt{\frac{r^2 r'^2}{R^2} + R^2 - 2 r r' \cos \alpha}} \quad (8.12)$$

where α is the angle between $\vec{r}$ and $\vec{r}'$. We then have

$$\left. \frac{\partial G_D}{\partial r'} \right|_{r'=R} = \frac{r^2 - R^2}{R (r^2 + R^2 - 2 R r \cos \alpha)^{3/2}} \quad (8.13)$$

so that

$$\begin{aligned}\phi(\vec{r}) &= \int G_D(\vec{r}, \vec{r}') \rho(\vec{r}') dV' \\ &+ \frac{1}{4\pi} \oint \phi(R, \theta', \varphi') \frac{(r^2 - R^2) R^2 \sin \theta' d\theta' d\varphi'}{R (r^2 + R^2 - 2 R r \cos \alpha)^{3/2}}\end{aligned}\quad (8.14)$$

where the first term vanishes if $\rho = 0$, and $\phi(R, \theta', \varphi')$ is the boundary condition on the sphere. Note that α depends on θ', φ' in the integral:

$$\begin{aligned}\cos \alpha = \frac{\vec{r}}{r} \cdot \frac{\vec{r}'}{r'} &= (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) \cdot (\sin \theta' \cos \varphi', \sin \theta' \sin \varphi', \cos \theta') \\ &= \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\varphi - \varphi')\end{aligned}\quad (8.15)$$

8.5.1 Expansion in spherical harmonics

More generally, for a given region we would want to write

$$G = \frac{1}{|\vec{x} - \vec{x}'|} + F \quad (8.16)$$

where the first term is to be written in spherical coordinates, and F solves the Laplace equation in spherical coordinates for the given boundary conditions

$$F = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} (A_{\ell m} r^{\ell} + B_{\ell m} r^{-(\ell+1)}) Y_{\ell m}(\theta, \varphi) \quad (8.17)$$

in terms of the spherical harmonics

$$Y_{\ell m}(\theta, \varphi) = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_{\ell}^m(\cos \theta) e^{im\varphi} \quad (8.18)$$

with P_{ℓ}^m the associated Legendre functions. For azimuthal symmetry, only $m = 0$ contributes

$$P_{\ell}^0(\cos \theta) = P_{\ell}(\cos \theta) \quad (8.19)$$

in terms of the Legendre polynomials

$$P_0(x) = 1 \quad , \quad P_1(x) = x \quad , \quad P_2(x) = \frac{1}{2} (3x^2 - 1) \quad , \quad \dots \quad (8.20)$$

These satisfy the orthogonality relations

$$\int_{-1}^1 P_{\ell'} P_{\ell} dx = \frac{2}{2\ell+1} \delta_{\ell'\ell} \Rightarrow \iint d\Omega Y_{\ell'm'}^* Y_{\ell m} = \delta_{\ell'\ell} \delta_{m'm} \quad (8.21)$$

One can show that

$$\begin{aligned} \frac{1}{|\vec{x} - \vec{x}'|} &= 4\pi \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{1}{2\ell+1} \frac{r_{<}^{\ell}}{r_{>}^{\ell+1}} Y_{\ell m}^*(\theta', \varphi') Y_{\ell m}(\theta, \varphi) \\ &= \sum_{\ell=0}^{\infty} \frac{r_{<}^{\ell}}{r_{>}^{\ell+1}} P_{\ell}(\cos \gamma) \end{aligned} \quad (8.22)$$

where γ is the angle between $\vec{x}'$ and $\vec{x}$. One then finds $A_{\ell m}$, $B_{\ell m}$ from the boundary conditions.

8.6 Example: a ring of charge

Consider a ring of total charge Q and radius a , lying in a horizontal plane at height b above the origin, as depicted in the Figure: each element of the ring sits at distance c from the origin, at polar angle α , with

$$c^2 = a^2 + b^2 \quad , \quad \tan \alpha = \frac{a}{b} \quad (8.23)$$

On the axis, the potential can be written directly

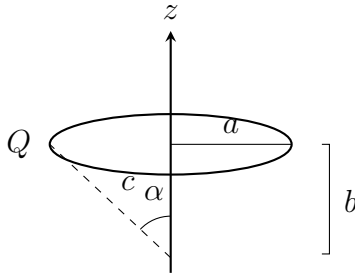


Figure 8.5: A ring of total charge Q and radius a in a plane at height b ; each element of the ring lies at distance $c = \sqrt{a^2 + b^2}$ from the origin, at polar angle α with $\tan \alpha = a/b$.

$$\phi(z) = \frac{Q}{(z^2 + c^2 - 2 z c \cos \alpha)^{1/2}} \quad (8.24)$$

Azimuthal symmetry implies $m = 0$ only. Expanding, for $z > c$

$$\phi(z) = Q \sum_{\ell=0}^{\infty} \frac{c^\ell}{z^{\ell+1}} P_\ell(\cos \alpha) \quad (8.25)$$

while for $z < c$

$$\phi(z) = Q \sum_{\ell=0}^{\infty} \frac{z^\ell}{c^{\ell+1}} P_\ell(\cos \alpha) \quad (8.26)$$

In general, we expect

$$\phi(r, \theta) = \sum_{\ell=0}^{\infty} (A_{\ell m} r^\ell + B_{\ell m} r^{-(\ell+1)}) \sqrt{\frac{2\ell+1}{4\pi}} P_\ell(\cos \theta) \quad (8.27)$$

and matching onto the on-axis expansion gives

$$\phi(r, \theta) = Q \sum_{\ell=0}^{\infty} \frac{r_{<}^\ell}{r_{>}^{\ell+1}} P_\ell(\cos \alpha) P_\ell(\cos \theta) \quad (8.28)$$

where $r_{<}$ ($r_{>}$) is the smaller (larger) of r and c .